

Philmont Geologic Synthesis for Trek 12-15

Executive summary

Philmont Scout Ranch sits on the eastern flank of the Cimarron Range, a Laramide uplift on the west edge of the Raton Basin. That setting is why the ranch packs so much geology into a relatively small area: Precambrian crystalline basement and Pennsylvanian red beds crop out near the mountain core, while Jurassic, Cretaceous, and Paleocene sedimentary rocks fringe the uplift to the east and north. Later igneous intrusion and volcanism overprinted that older section, especially around Baldy Mountain and the central-southern landmark intrusions such as the Tooth of Time. ¹

For an advanced hiker, the practical takeaway is simple. On Trek 12-15, you are not walking through one uniform “Philmont geology.” You are moving from basin-fill and coastal-plain strata, into mesa-forming volcanic and intrusive terrain, up into the Baldy intrusive-mining district, and then back down through Cretaceous–Paleocene clastic rocks and sandstone cliff country that preserves archaeology and dinosaur-track context. The itinerary itself confirms that progression: Turkey Creek Trailhead to Harlan, Devils Wash Basin, Cimarron River, Santa Claus, Baldy Town, Pueblano, Elkhorn, Ponil, Indian Writings, and finally Six Mile Gate. ²

USGS Professional Paper 505 remains useful because its field explanations are still broadly right: Philmont’s scenery reflects a sedimentary section deformed at the edge of the Rockies and cut by younger igneous bodies. Where it is dated is in stratigraphic detail and regional interpretation. Modern work has refined the Triassic and Lower Cretaceous nomenclature in northeastern New Mexico, sharpened the Raton Basin’s Late Cretaceous–Paleocene framework, tied younger volcanism to the broader Raton-Clayton volcanic field and Jemez Lineament context, and added discoveries unknown in 1965, including the Philmont tyrannosaur track and later K–Pg boundary work from the broader basin. ²

For Trek 12-15 specifically, the most educational part of the route is the Baldy–French Henry interval. That segment lets you see the difference between host sedimentary rocks and younger intrusions, the structural control of veins and mine workings, the erosional expression of harder intrusive rocks, and the way mining followed veins, alteration, and accessible adits rather than just topography. The second most educational part is the Indian Writings to Six Mile Gate exit, where resistant sandstone cliff and ledge-forming units create the kind of durable exposed surfaces that preserve both rock art settings and dinosaur-track localities in the broader North Ponil area. ³

A key limitation remains: the itinerary page you provided gives route order, mileage, and camps, but not exact camp coordinates or UTM grid references. Exact formation-by-formation crossing points therefore require either the Philmont North and South sectional maps, or a GPS/KMZ/GPX overlay registered to the regional geologic map. In the day-by-day section below, I therefore distinguish between **high-confidence** geology and **probable route inferences**. ³

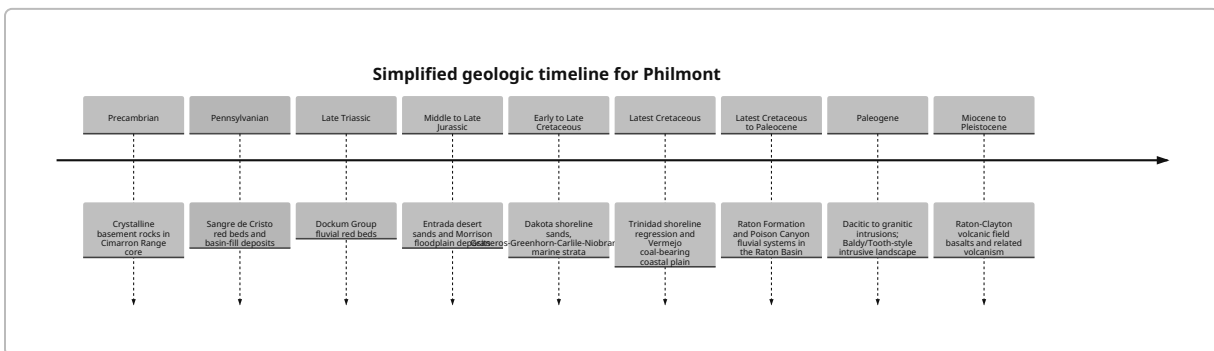
Regional framework

Philmont lies where the Southern Rockies meet the western Great Plains. Structurally, the Cimarron Range is a Laramide uplift, with the Raton Basin immediately adjacent. The regional geology is therefore a classic foreland-margin story: older basement and Paleozoic rocks were uplifted, Mesozoic and latest Cretaceous–Paleocene strata accumulated in the adjacent basin, and younger igneous events intruded and locally erupted through that package. 4

The broad chronology is well established. The oldest rocks relevant to Philmont are Precambrian crystalline rocks in the mountain core; over them lie Pennsylvanian red beds of the Sangre de Cristo basin-fill system; then Triassic fluvial red beds of the Dockum Group; Jurassic eolian and fluvial units including Entrada and Morrison; Lower and Upper Cretaceous shoreline-to-marine units as the Western Interior Seaway advanced and retreated; then Maastrichtian to Paleocene coal-bearing and fluvial strata of the Trinidad, Vermejo, Raton, and Poison Canyon succession in the Raton Basin. Younger Tertiary intrusions and later basaltic volcanism completed the picture. 5

Modern work differs from PP 505 in three especially important ways. First, northeast New Mexico's Triassic and Lower Cretaceous section is now split and correlated more precisely than the older “lumped” terminology common in mid-century field guides; the Dockum Group, for example, is now commonly discussed in terms of formations such as Baldy Hill, Travesser, Sloan Canyon, and Sheep Pen, and Lower Cretaceous units may be treated more precisely than simply “Dakota.” Second, the Raton Basin's latest Cretaceous and Paleocene section is better integrated with sequence-stratigraphic and basin-evolution studies. Third, younger volcanism is now framed within the Raton-Clayton volcanic field and Jemez Lineament context, a modern tectonomagmatic interpretation not available to 1965 authors. 6

This simplified timeline synthesizes that framework. 7



Field-scale tectonic reading at Philmont follows from that history. The eastern ranch presents basin-margin sedimentary outcrop belts. The northwestern ranch exposes the intrusive Baldy district. Linear ridges and abrupt resistant spurs commonly reflect sandstone ledges, dikes, sills, or intrusive contacts; broad, gentler slopes often reflect shale- or mudstone-rich units. When you move from yellow-orange, micaceous, cliffy to ledgy clastics into obviously crystalline or porphyritic rock with quartz veins and mine debris, you have crossed from basin stratigraphy into the intrusive-mineralized core of the district. 8

Stratigraphy and field recognition

The most useful field-booklet format is not a full formal stratigraphic column, but a “what does this look like on the trail?” table. The table below emphasizes the units most likely to matter on Trek 12-15, either because the route probably crosses them or because they explain the scenery you will repeatedly see from overlooks. Exact crossing points are in several cases route inferences from your itinerary plus the regional geologic framework, not survey-grade contact picks. [filecite turn0file0](#)

Formation or unit	Approximate age	Typical lithology	Fast field ID	Likely relevance on Trek 12-15	Source basis
Precambrian crystalline basement	Precambrian	Gneiss, schist, granodiorite	Crystalline, foliated or coarse-grained, no bedding	Most relevant as regional core beneath Baldy district; likely exposed or closely approached in the upland intrusive/core zone	Regional framework from Cimarron uplift sources. 4
Sangre de Cristo red beds	Pennsylvanian	Red sandstone and conglomerate	Brick-red color, coarse clasts, thick beds	More important regionally than as a certain on-route unit; may contribute to older sedimentary section near uplift margins	Regional stack synthesis. 4
Dockum Group	Late Triassic	Red beds: sandstone, siltstone, mudstone	Red-purple mudstone, fluvial sandstone lenses, badland weathering	Mostly northern/northeastern regional context; not a high-confidence day-by-day unit on this itinerary, but relevant to the broader Philmont region	9

Formation or unit	Approximate age	Typical lithology	Fast field ID	Likely relevance on Trek 12-15	Source basis
Entrada Sandstone	Middle Jurassic	Sandstone, locally eolian	Smoother, rounded, cross-bedded sandstone; cliff to slope-forming	Possible in cliff/ledge country east of the intrusive core, but confidence is moderate without coordinates	10
Morrison Formation	Late Jurassic	Mudstone, sandstone, siltstone, limestone	Variegated green-gray-red mudstone, softer slopes under resistant sandstones	Probable in some lower-elevation sedimentary sections east/northeast of the intrusive zone	11
Dakota and related Lower Cretaceous shoreline sands	Early to early Late Cretaceous	Quartz sandstone with shale/siltstone interbeds	Yellow-gray quartz sandstone, crossbedding, ledgy cliffs, pebbly horizons	High relevance for Indian Writings cliff country and some eastern/northern ledges	12
Graneros–Greenhorn–Carlile interval	Late Cretaceous	Shale with limestone/sandstone beds	Dark gray shale slopes with harder limestone ledges/concretions	Likely contributes to broad slope-and-bench terrain in the basin-margin part of the trek	13
Niobrara–Pierre interval	Late Cretaceous	Chalky limestone to dark marine shale	Gray fissile shale or lighter carbonate beds; marine concretions possible	Likely in northern/eastern basin-margin country, especially where ledge-over-slope topography appears	14

Formation or unit	Approximate age	Typical lithology	Fast field ID	Likely relevance on Trek 12-15	Source basis
Trinidad Sandstone	Maastrichtian	Fine- to very fine-grained arkosic sandstone	Massive shoreface sandstone, resistant ledges or cliffs	Good candidate for resistant sandstone cliffs in northern Philmont country, including rock-art settings	15
Vermejo Formation	Maastrichtian	Sandstone, shale, carbonaceous shale, coal	Softer gray beds, carbonaceous material, thin coal seams, less cliffy than Trinidad	Highly relevant in the northern basin section around Ponil country	16
Raton Formation	Latest Cretaceous to Paleocene	Sandstone, shale, coal, basal conglomerate	Gray to buff sandstone, coal/ carbonaceous beds, local conglomerate	Very relevant to northern Philmont and dinosaur-track context in the broader North Ponil area	17
Poison Canyon Formation	Latest Cretaceous to Paleocene	Arkosic sandstone, mudstone, siltstone, conglomerate	Yellow-orange micaceous sandstone, coarse channel bodies, non-coaly	Very likely on central-to-northern trek segments; a good candidate for yellow-orange slope-and-cliff terrain	18
Tertiary intrusive rocks	Paleogene	Dacite porphyry to granitic intrusive rocks	Porphyritic texture, hard gray-pink rock, massive outcrops, baked contacts	Dominant in Baldy district and in iconic intrusive landforms such as the Tooth of Time	19

Formation or unit	Approximate age	Typical lithology	Fast field ID	Likely relevance on Trek 12-15	Source basis
Basalt and related volcanic rocks	Miocene to Pleistocene	Basalt flows, cinder-cone products, volcanic breccia	Dark dense rock, vesicles, resistant mesa caps or rubbly float	Most important as regional volcanic context and probable mesa cap influence near central/northeastern high ground	20

In practice, hikers can simplify the route into four repeated field patterns. **Pattern one** is resistant quartzose sandstone ledges and cliffs: likely Dakota-related or Trinidad-type units. **Pattern two** is softer gray or yellow slope formers: likely shale, mudstone, or weakly cemented fluvial units in the Vermejo, Raton, Poison Canyon, or marine shale intervals. **Pattern three** is dark rubbly lava or vesicular volcanic float on high benches or mesas. **Pattern four** is hard crystalline or porphyritic intrusive rock with quartz veining and mine debris in the Baldy district. 21

Field identification should focus less on exact formal names and more on what you can actually test. Quartz-rich sandstones will feel gritty and commonly preserve cross-bedding and sharp ledges. Carbonaceous or coal-bearing beds are dark, dirty, and friable. Marine shales weather to slopes and often carry concretions or thin limestones. Porphyritic intrusions show a speckled “two-size” texture, with larger crystals in a finer matrix. Basalt is dark, dense, and commonly vesicular. Vein quartz is milky white and much harder than the host rock. 22

Magmatism, structure, mining, fossils, and archaeology

Philmont’s most famous skyline features are not random erosional oddities; they are expressions of younger igneous intrusion and differential erosion. The Tooth of Time is widely described as an intrusive dacite-porphyry body that remained standing after the surrounding sedimentary rocks weathered away. That same general intrusive story scales up into the Baldy district, where younger igneous bodies and associated hydrothermal systems overprinted the older stratigraphic section. 19

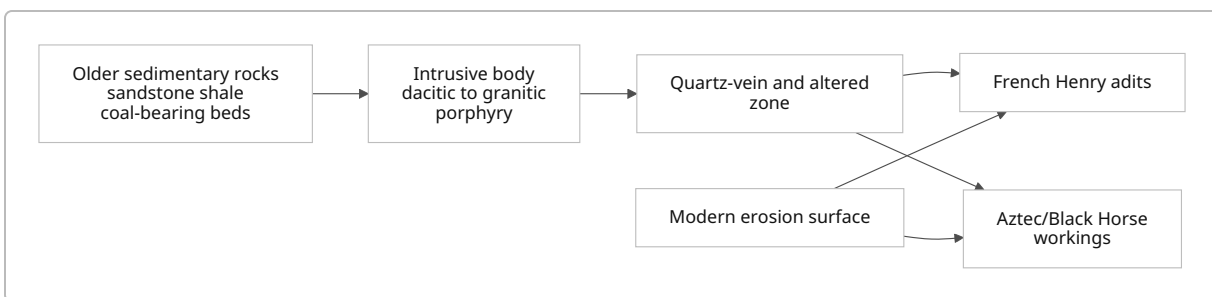
Structurally, the ranch sits on a basin-margin uplift. At regional scale, the Cimarron Range is bounded by major faults associated with the Laramide uplift, while the adjacent Raton Basin preserves a thick asymmetric sedimentary section. At field scale, the most important structural indicators for hikers are simpler: repeated straight-line ridges, joint-controlled cliff faces, dike-like resistant ribs, abrupt lithologic breaks, and mine adits or dumps aligned along vein trends or contact zones. In the Baldy area especially, the landscape itself often telegraphs where rock competence changes and where miners expected ore to persist. 23

The Baldy Mining District became one of New Mexico’s major gold districts after gold discoveries in 1866. The district included at least the Aztec, French Henry, Black Horse, Mystic, Bull-of-the-Woods, Rebel Chief, and other mines, and historically produced chiefly gold, with silver and copper also important in parts of the district. The district’s ore was worked through an evolving infrastructure of stamp mills, tramways, adits,

shafts, and later reprocessing efforts, but the basic mining logic stayed the same: chase quartz-rich or otherwise mineralized structures where grade and access justified it. 24

The most cautious interpretation of the ore controls, given the available source base here, is that Baldy and French Henry are intrusion-related vein systems hosted in or adjacent to sedimentary rocks and intrusive bodies, with quartz-rich lodes carrying gold and locally silver or copper. That is a district-scale inference supported by the repeated historical references to quartz-rich veins, intrusive rocks, hydrothermal-style mineralization, and mine development clustered around the intrusive Baldy core. It is safer than claiming a single deposit model for every mine, because the district clearly included more than one ore style and more than one commodity. 25

This simplified cross-section is the right mental model for the Baldy–French Henry area. It is schematic, not to scale. The intent is to show the relationship between older basin strata, younger intrusive rock, structurally controlled veins, and the mine levels developed to follow those veins. The geometry is a district-scale teaching tool rather than a formal mine map. 26



A concise mining dossier helps separate what is historically secure from what is interpretive. Gold was found in the district in 1866; Baldy Town developed as the service center for east-side operations; the Aztec became the main producer on that side of the mountain; French Henry opened in 1870, revived in 1894, and ultimately ceased by 1938; the district's final decades were dominated by repeated attempts to rediscover richer lodes or rework old material, including the well-known Deep Tunnel effort and later mill reconfiguration. Observable remnants today include road and tramway grades, dumps and tailings, collapsed or barred portals, mill-site remains, cabin or foundation remains, and localized alteration/vein float. 27

Paleontology adds a very different layer to the trek. The Philmont tyrannosaur footprint, formally named *Tyrannosauripus pillmorei*, was discovered by Charles Pillmore and described in 1994. It is widely presented as the only documented *T. rex* track, but the exact discovery year is inconsistent across online summaries: some sources say 1983, others 1993. The safer statement is that the track was discovered by Pillmore in the 1980s or early 1990s, formally described in 1994, and comes from latest Cretaceous track-bearing strata in the North Ponil area. 28

For hikers, the lesson is not just “a dinosaur walked here.” It is that latest Cretaceous northern Philmont preserved wet, muddy, low-relief depositional surfaces capable of taking and preserving large theropod tracks, and that those surfaces are part of a larger Raton Basin section that also records shoreline retreat, floodplains, coal swamps, and the end-Cretaceous transition. That is a major advance over PP 505 simply because the track had not yet been found when the report was published. 29

Archaeology at Indian Writings is tied to geology in a quieter but equally real way. The itinerary confirms an Ancestral Puebloan archaeology program there, and the northern Philmont landscape includes resistant sandstone cliff and shelter-forming units capable of preserving broad exposed rock faces. The most defensible geologic reading is that Indian Writings exists where durable ledge- or cliff-forming sandstone provides a stable writing surface, visual prominence, and shelter geometry in an otherwise erosive canyon system. Exact bed assignment at the site needs a georeferenced sectional-map overlay.

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Trek analysis

The analysis below is built from your itinerary route order and mileage. filecite turn0file Exact UTM contact points are **not** available from that page alone, so where I name formations for a day, I am distinguishing **high-confidence units** from **probable** ones inferred from the regional outcrop belts and the Baldy-centered geography of the route. 31

Days through Harlan and Devils Wash Basin

The first hiking day to Harlan and the next day to Devils Wash Basin likely traverse the central-to-northern basin-margin section east of the Baldy intrusive core. The most likely dominant rocks are Late Cretaceous to Paleocene clastics of the Raton Basin, especially Raton- and Poison-Canyon-type sandstones, siltstones, and mudstones, with local volcanic influence on mesa surfaces or nearby high ground. Expect bench-and-slope topography, yellow-orange to gray weathering clastics, and local ledgy resistant beds rather than the obviously mineralized intrusive textures that will appear later. filecite turn0file 32

What to watch for in hand sample and outcrop: coarse or micaceous sandstone that breaks into gritty blocks; softer mudstone or shale slopes; isolated channel-form sandstone bodies standing out from surrounding slopes; and any change from gray, carbonaceous beds into yellow-orange arkosic sandstone, which is often the practical field clue that you have moved from Raton-like into Poison-Canyon-like facies.

33

The best teaching points for these days are that basin stratigraphy controls trail feel, that coarse channel sand bodies form ledges and benches while muddy floodplain beds form slopes, that “mesa country” here may reflect either resistant sandstone or younger volcanic caprock, and that this pre-Baldy terrain sets up the contrast that makes the intrusive mining district stand out so strongly later in the trek. 34

Days through Cimarron River and Santa Claus

The descent to Cimarron River and then climb to Santa Claus likely continues across the mixed sedimentary section, but with stronger geomorphic contrast because river incision exposes more of the vertical stack. This is the point in the itinerary where you should expect topography to tell you as much as lithology: canyon bottoms in softer units, steeper ledges where resistant sandstone is present, and topographic breaks that probably align with formation changes. filecite turn0file 35

What hikers should observe is the relationship between stream incision and rock strength. Look for ledgy, blocky, more resistant sandstone around constrictions; finer, recessive slopes where mudstone or shale dominates; and any carbonaceous or dark-gray beds that hint at coal-bearing or swampy depositional

environments in the broader Vermejo–Raton interval. These are the days to practice reading landforms before the dramatic intrusive terrain makes the geology seem more obvious than it really is. 36

Teaching points here: the Cimarron drainage is an erosional cross-section through the basin margin; sandstone formers make cliffs and steps; softer shale-rich packages make broader, brushier slopes; and the alternation of cliffs and recesses is not random scenery but a direct expression of stratigraphic competence contrasts. 37

Day to Baldy Town through Head of Dean

The long day to Baldy Town is where the geology turns from basin stratigraphy into upland intrusive-mining country. By the time you approach Head of Dean and Baldy Town, you should expect to encounter the strongest evidence yet of Tertiary intrusion: harder massive rock, porphyritic textures, quartz-vein float, altered ground, mining debris, and more abrupt relief. This is the geologic hinge point of Trek 12-15.

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Hand-sample and outcrop tips: start distinguishing sedimentary bedding from intrusive fabric. Bedded sandstones show planar layering and sorting; intrusive porphyry looks massive and speckled, with larger crystals in a finer matrix. White vein quartz or iron-stained fracture fillings are major clues that you are entering hydrothermal terrain. Trail cobbles with milky quartz, rusty coatings, or visibly mixed lithologies are not incidental here. 39

Educational talking points: why ore and mines cluster where they do; why Baldy's scenery is more rugged than adjacent basin country; why intrusive rocks resist erosion differently from shale and coal-bearing strata; and why mining infrastructure followed vein geometry and processing logistics rather than merely the easiest walking route. 24

Baldy layover and summit day

The Baldy layover is the geologic centerpiece of the trek. Baldy is the highest point on the ranch and sits in the core of the mining district, where intrusive rock, structural control, relief, and mining history all coincide. The Black Horse Mine and summit traverse make this the best day to teach the difference between regional tectonic setting and local ore geology. filecite turn0file 40

At outcrop scale, expect massive hard rock, fracture networks, vein material, mine waste, and altered zones. At landscape scale, look west and think uplifted mountain core; look east and think basin stratigraphy stepping down into the Raton Basin. When the weather is clear, Baldy is one of the best places on Philmont to talk about why a Laramide uplift produces both dramatic relief and a thick adjacent basin fill. 41

The best educational points are these: the mountain is not just "high," it is structurally and lithologically different; miners targeted veins and lodes created by post-sedimentary magmatism and hydrothermal circulation; a summit panorama is a geologic map in three dimensions; and Baldy's relief is a product of both uplift and selective erosion of hard rock. 42

Day from Baldy Town to Pueblano through French Henry

This is the most mining-specific day on the itinerary. You pass the Claude Mining and Milling Company program at French Henry and then continue toward Pueblano. Geologically, this is your best chance to see how a mine, a mill site, access routes, and the surrounding ridges all relate to the same vein-controlled district.   43

What to observe: dumps and waste rock compared with natural scree; tailings or reworked mill-site sediment versus stream alluvium; obvious vein quartz versus ordinary sandstone or conglomerate clasts; and the difference between the inaccessible mine proper and the more visible processing/remnant landscape around the historic mill site. If you see iron staining, silicified float, angular quartz-rich debris, and old worked ground in the same place, read the whole scene as ore plus access plus processing. 44

Teaching points: ore grade and milling are separate problems; many mines fail not because ore is absent but because concentration, transport, water, or capital are inadequate; French Henry's history is a good case study in repeated optimism followed by metallurgical and economic disappointment; and visible historical remains are themselves geologic evidence because they show where rock was worth moving by hand, burro, tramway, or mill. 45

Day to Elkhorn

The short move to Elkhorn is a transition day out of the highest-intensity mining district and back into more mixed geomorphic country. Depending on exact route location, you are probably leaving obviously intrusive-mineralized terrain and re-entering the broader sedimentary outcrop belts that control northern Philmont's canyon-and-bench topography. This is where it helps to ask what changes underfoot: less quartz-vein float, fewer mine indicators, more bedded clastics, and a greater return to landforms governed by sandstone-versus-shale contrast.   46

Educational points: how quickly the mining signal fades once you leave the intrusive core; how the same drainage network can cross radically different rock packages; why trail construction in this kind of terrain must account for shale slopes versus sandstone ledges; and how vegetation shifts often track substrate and soil changes that hikers first notice as "the trail feels different." 47

Day to Ponil

The route to Ponil likely runs through northern basin-margin sedimentary country dominated by Cretaceous to Paleocene clastics, with the exact mix depending on where the trail sits relative to the Trinidad–Vermejo–Raton–Poison Canyon belts. Ponil's scenic quality owes a lot to repeated resistant sandstone bodies framing drainages cut into softer beds.   48

Field ID here should emphasize color and resistance. Massive ledgy sandstone is the easiest unit to spot from a distance. Gray, carbonaceous, less-resistant beds suggest Vermejo or Raton tendencies. Yellow-orange, coarser, more arkosic sandstone suggests Poison Canyon tendencies. Even if you cannot assign every exposure formally, you can almost always place it into the right depositional style: shoreline sand, swampy coal-bearing plain, or younger coarse fluvial channel system. 49

Teaching points: final retreat of the Western Interior Seaway is recorded in the Trinidad Sandstone; coal-bearing coastal plain conditions are recorded in Vermejo and parts of the Raton section; younger fluvial reworking is recorded in Poison Canyon; and Ponil canyon scenery is the erosional expression of that stacked history. ⁴⁸

Day to Indian Writings over Hart Peak

This is one of the most interpretively rich days of the trek because geology and archaeology converge. The itinerary explicitly ties Indian Writings to an Ancestral Puebloan archaeology program, and the landscape strongly suggests resistant sandstone cliffs and ledges providing durable exposed rock faces. The most plausible on-trail teaching framework is that people used a geologically favorable place: prominent, accessible, sheltering, and visually persistent. ³⁰

What hikers should observe is not only the carvings themselves but the rock surface quality: hardness, grain size, weathering rind, visibility from below, and the way the cliff or shelter sits within the canyon. A good field question is, “Why here and not fifty yards away?” The answer is usually geological first and cultural second: the right face had to exist before it could be chosen for repeated human use. ⁵⁰

Talking points: resistant sandstone controls cliff geometry; stable outcrop surfaces preserve both natural textures and cultural markings; canyons concentrate travel routes and visibility; and archaeology at Philmont is inseparable from geomorphology because rock type controls what can survive. ³⁰

Exit day to Camping Headquarters through the T-Rex Track area

The final day back to headquarters includes the T-Rex Track stop noted in your itinerary. That makes this day disproportionately significant geologically, even though the hiking mileage is short. You are moving through the same broader northern Philmont sedimentary setting that preserves latest Cretaceous track-bearing surfaces in the North Ponil area. ⁵¹

What to look for at any track-bearing or track-explanatory stop is the sedimentary story: fine muddy or silty substrate capable of recording a print, rapid enough burial to protect it, and later exhumation without total destruction. Even if the exact visited track display is not in pristine in-place form, the key lesson is that dinosaur tracks require the right substrate and the right burial history, not merely the presence of dinosaurs. ⁵²

Talking points: Philmont preserves more than “rocks” because depositional surfaces matter; latest Cretaceous landscapes here included low-gradient, wet, trackable surfaces; track preservation is a sedimentological event as much as a paleontological one; and this stop is one of the clearest examples of how modern discoveries added to the older PP 505 story rather than replacing it. ⁵³

Recommended sources, figure ideas, and quick field appendix

The most useful source stack for converting this report into a field booklet is a combination of classic USGS interpretation, one good regional map, one trail-oriented guide, and a few modern stratigraphic and mining references. PP 505 is still the best narrative introduction; the Philmont regional geologic map is the best

spatial control; the old trail guide remains useful for field stops; and modern northeastern New Mexico and Raton Basin papers are what you use to update nomenclature and interpretation. 54

A practical “get these next” list is below. The URLs are formatted in code so you can copy them easily.

USGS Professional Paper 505
`https://pubs.usgs.gov/pp/505/`

Direct PDF version you already referenced
`https://pubs.usgs.gov/pp/pp_505/pdf/pp_505.pdf`

Geologic Guide to the Trails at Philmont Scout Ranch, OFR 22
`https://geoinfo.nmt.edu/publications/openfile/details.cfm?Volume=22`
`https://geoinfo.nmt.edu/publications/openfile/downloads/0-99/22/ofr_22.pdf`

USGS Geologic Unit Lexicon entry for Dockum Group
`https://ngmdb.usgs.gov/Geolex/NewUnits/unit_7944.html`

USGS Raton Basin K-Pg guide via archive reference trail
`https://esp.cr.usgs.gov/content/sequences/sequence-home`

NMBGMR / New Mexico Geological Society modern regional review
Search title:
“Geology of Northeastern New Mexico, Union and Colfax Counties, New Mexico”

For a field booklet, the best figures are not generic textbook figures but route-specific ones. The most valuable set would be: a geologic map overlay showing the Trek 12-15 line against the regional formations; a simplified elevation profile annotated by rock type; a Baldy summit panorama labeled with intrusive core versus basin strata; a French Henry schematic showing mine, mill, dumps, and drainage; an Indian Writings spread explaining why the cliff exists geologically; and a final North Ponil/T-Rex-track page showing how a muddy substrate preserves a large track. Those figure ideas follow directly from the itinerary and the geology summarized above. 55

The most important data gaps are straightforward. Exact formation crossings and UTM coordinates for your route are still missing because the itinerary sheet is not georeferenced. To close that gap, obtain the Philmont North and South sectional maps and, ideally, your trek line as GPX/KMZ. Once those are registered to the Philmont regional geologic map, the day-by-day discussion above can be upgraded from “probable formation belts” to “exact mapped contacts crossed at these milepoints.”

For the back-of-book appendix, keep the field key brutally simple:

Quick key	Use in the field
Quartz sandstone	Gritty, hard, often cliff-forming, usually light gray/yellow/tan

Quick key	Use in the field
Carbonaceous shale/coal-bearing beds	Dark gray to black, dirty, soft, commonly slope-forming
Marine shale	Gray fissile slope former, may include concretions or thin limestones
Arkosic sandstone/conglomerate	Coarser, feldspathic, often yellow-orange, fluvial look
Porphyritic intrusive rock	Massive, hard, speckled texture with larger crystals in fine matrix
Basalt	Dark dense rock, may show vesicles
Vein quartz	Milky white, glassy to sugary, much harder than host rock
Iron staining	Rusty orange-red coatings along fractures or dumps; often worth noting in mining terrain

Those identifications reflect the dominant rock styles actually relevant to Trek 12-15: sandstone ledges, shale slopes, coarse fluvial clastics, intrusive porphyry, basaltic float, and quartz-vein material in the Baldy district. ⁵⁶

A compact hardness and fossil note is enough for hiking use. Mohs benchmarks worth memorizing are fingernail about 2.5, copper coin about 3, knife blade or glass around 5–5.5, and quartz 7. For fossils, the most realistic checklist on this route is not body fossils you will casually find on trail but **contexts**: coal-bearing or plant-rich beds in the Vermejo–Raton interval, marine invertebrate potential in the Cretaceous marine units, and track-bearing surfaces in latest Cretaceous northern Philmont. ⁵⁷

The blunt bottom line is this: **PP 505 is still a good field companion, but it is no longer enough by itself.** For Trek 12-15, the best modern booklet will use PP 505 for readable explanations, the geologic map for spatial control, modern northeastern New Mexico stratigraphy for updated names and ages, and the Baldy/track/Indian Writings stops for route-specific interpretation. That combination will give you something far better than a generic geology handout: a field guide you can actually use while hiking. ⁵⁸

¹ ⁴ ²³ ²⁶ ³¹ ⁴¹ ⁴² https://en.wikipedia.org/wiki/Cimarron_Range%2C_New_Mexico

https://en.wikipedia.org/wiki/Cimarron_Range%2C_New_Mexico

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https://en.wikipedia.org/wiki/Poison_Canyon_Formation

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- 13 https://en.wikipedia.org/wiki/Graneros_Shale
https://en.wikipedia.org/wiki/Graneros_Shale
- 14 https://en.wikipedia.org/wiki/Niobrara_Formation
https://en.wikipedia.org/wiki/Niobrara_Formation
- 15 22 37 48 49 56 https://en.wikipedia.org/wiki/Trinidad_Sandstone
https://en.wikipedia.org/wiki/Trinidad_Sandstone
- 16 36 57 https://en.wikipedia.org/wiki/Vermejo_Formation
https://en.wikipedia.org/wiki/Vermejo_Formation
- 17 32 33 https://en.wikipedia.org/wiki/Raton_Formation
https://en.wikipedia.org/wiki/Raton_Formation
- 19 39 https://en.wikipedia.org/wiki/Tooth_of_Time
https://en.wikipedia.org/wiki/Tooth_of_Time
- 20 https://en.wikipedia.org/wiki/Raton-Clayton_volcanic_field
https://en.wikipedia.org/wiki/Raton-Clayton_volcanic_field
- 25 https://en.wikipedia.org/wiki/Baldy_Town%2C_New_Mexico
https://en.wikipedia.org/wiki/Baldy_Town%2C_New_Mexico
- 28 29 51 52 53 <https://en.wikipedia.org/wiki/Tyrannosauripus>
<https://en.wikipedia.org/wiki/Tyrannosauripus>
- 30 50 54 58 https://en.wikipedia.org/wiki/Philmont_Scout_Ranch
https://en.wikipedia.org/wiki/Philmont_Scout_Ranch
- 40 https://en.wikipedia.org/wiki/Baldy_Mountain_%28Colfax_County%2C_New_Mexico%29
https://en.wikipedia.org/wiki/Baldy_Mountain_%28Colfax_County%2C_New_Mexico%29
- 43 44 45 55 https://en.wikipedia.org/wiki/French_Henry_mine
https://en.wikipedia.org/wiki/French_Henry_mine